

capability does not imply detection capability at the pragmatic level; the model can produce ironic text by following a register-pragmatic prompt without learning to recognize the implicature from the surface form alone.

4. DISCUSSION

4.1. Model capability mirrors detection capability

The three generators show a consistent ordering — DeepSeek > Gemma3 > Qwen3 — in both generation authenticity and detection performance (Section 3.5). Qwen3, the cheapest local model, fails both tasks: it cannot *generate* authentic krama Javanese (defaulting to Indonesian), and it cannot *detect* krama hate. Gemma3 partially succeeds at generation (56%) and detection (67% overall), while DeepSeek leads both. This parallelism **may reflect** a shared underlying competence — models that understand Javanese register well enough to generate it may also understand it well enough to detect it. **We offer this as an observation to be tested rather than an inference, for two reasons: the pattern is correlational across only three models, and it does not hold uniformly within a model. DeepSeek generates *krama-sarcastic* hate to an explicit specification yet detects only 11% of it (Section 3.5), a dissociation that a single shared competence would not predict.**

An additional finding: Qwen3 hallucinated the word *kacandran* (non-existent in standard Javanese; likely a confabulation of *kekacauan* with Javanese morphology) in five of nine ngoko-direct examples. This is a concrete instance of a known LLM failure mode — lexical confabulation in a low-resource language — and underscores the necessity of native validation even for the "easiest" niche.

4.2. The authentic-but-evasive cross-tab

Combining native authenticity (Section 3.2) with the finer-grained JELAS_HATE judgment collected from Yekti and Daniel (Section 2.7, Section 3.4) refines this cross-tabulation. Of the 9 cells that evade all 5 detectors (Section 3.5), neither Yekti nor Daniel independently marks any of them JELAS_HATE=1 (0/9 for both) — but the reasons are not uniform, and collapsing them into a single 0/9 headline would overstate the case. Classifying each cell by its validators' notes: 3 of the 9 target *politik_kolektif*, and both validators explicitly describe these as political cynicism rather than a SARA-identity attack — a scope question about whether "political collective" belongs alongside ethnicity/religion/gender as "identity hate" (Section 4.6, Limitation 6), not a pragmatic-blindness finding. One cell (intra-Java *Arek-vs-Mataraman* sarcasm) leaves the attacked party unstated, which both validators flag as a construction-clarity problem rather than an irony-recognition one. The remaining **5 of the 9** — targeting *suku_tionghoa*, *agama_islam*, *agama_kristen*, *gender_lgbtq*, and *suku_arab* — are described by both validators as "*ironi samar*" (ambiguous irony) or "deniable," several explicitly noting the text reads as sincere praise. These five cells most directly support this paper's blind-spot thesis: independent native and near-native readers, given the intended pragmatic mechanism in advance, still cannot confidently call the text hate — the same plausible deniability that makes *pasemon* an effective face-threat-mitigation strategy in Javanese also resists confident labeling, by machine or human. We therefore narrow the claim accordingly: automated detectors and human raters converge on uncertainty for the same 5/9 *krama-sarcastic* SARA-target stimuli, which is evidence of genuine pragmatic ambiguity rather than simple detector failure on otherwise-obvious hate. The remaining 4 evasive cells reflect a separate, disclosed limitation in the *politik_kolektif* target category and in one under-specified stimulus (Section 4.6, Limitation 6), and are not folded into the blind-spot claim.

4.3. Instrument and sociolinguistic effects in validator disagreement

Why do Mukhlis and Yekti disagree? What drives the divergence is not fully an open question: a follow-up analysis of the disagreement rows themselves finds a specific, largely explicable pattern,

rather than unstructured noise. All 39 Mukhlis–Yekti disagreements run in one direction (Mukhlis judged "not authentic," Yekti judged "authentic"); 19 of the 27 *krama-sarcastic* items (70% of that niche) are disagreement rows, and the niche accounts for 19/39 (49%) of all disagreements — either way, *krama-sarcastic* (already the hardest niche in Table 5) dominates. In 34 of the 39 rows Yekti's own notes mark the item as ambiguous on the hate dimension (JELAS_HATE = 0), 31 of which explicitly cite code-mixing with Indonesian as unremarkable to her ("*Campur wajar*" — mixing is normal).

Harmonizing raises Mukhlis–Yekti α from 0.095 to 0.519 (Table 7) — moderate, not chance-level. This correction does not extend to Daniel, whose agreement with Mukhlis instead *drops* under the same transformation, and whose 12 disagreements pair JELAS_HATE = 1 with OTENTIK = 0 in 9 cases, consistent with a non-native rater doubting register correctness even when the hateful content is legible to him.

Two things follow. First, a real part of the apparent native–native disagreement is an instrument artifact: Yekti's and Daniel's form separates "is this real Javanese" from "does this register as hate," while Mukhlis's original single-column form (fielded before the two-dimension split was introduced) could not. Second, the remaining, non-artifactual part tracks validator linguistic background: Yekti's home environment normalizes Javanese–Indonesian code-switching as everyday speech, while the first author's more Javanese-homogeneous environment treats the same code-mixing as a purity failure — a distinction consistent with documented urbanization- and class-linked *krama* attrition in Javanese sociolinguistics [30]. This is the same axis as Limitation (4) (Section 4.6) below (Central-Javanese prestige-purity norms vs. heterogeneous/Arek-urban norms), surfacing on the validation side of the pipeline rather than the generation side.

A related, unresolved question this raises: "authentic Javanese" as judged here reflects the specific linguistic communities of our three validators, and content targeting an ethnic-minority group with its own anecdotally reported, distinct Javanese sociolect (e.g., Chinese-Indonesian speakers, whose Javanese is reportedly distinguishable to other native ears) was not separately checked by a rater from that community — an open question for future validation rounds, not a claim this paper makes.

Future authenticity claims for this stimulus-construction method should still report a range or per-rater breakdown rather than a single pooled percentage.

4.4. Comparison with prior work

The finding that pragmatic implicature defeats detection aligns with, but is not reducible to, the English-language implicit-hate literature. Latent Hatred showed that implicit hate degrades classifier performance even with dedicated benchmarks [18], and ToxiGen demonstrated that large-scale machine generation can produce implicitly toxic statements that fool classifiers [19]. Both lines of work, however, operate in a monolingual, register-flat setting: the failure they document is the absence of surface toxicity. The Javanese blind spot documented here is structurally different — it is carried by a grammaticalized honorific register whose polite surface is systematically misread as benign, a mechanism unavailable in English. Functional-test resources have not yet reached this axis: Multilingual HateCheck spans ten languages without any Indonesian regional language [21], and SEAHateCheck, the closest contemporaneous work, covers four Southeast Asian national languages but not Javanese [20]. On the construction side, generated hate speech has been used to fine-tune and stress-test detectors [15], and zero-shot LLM detection is known to degrade across languages [44]; this paper couples those two threads to a register-stratified design in which generation makes an otherwise-uncollectable register testable, and native validation grounds the stimuli in a specific speech community.

4.5. Contribution to informatics and computer science

Beyond its sociolinguistic interest, this work makes three contributions that belong to informatics rather than to descriptive linguistics or moderation policy. First, it identifies a *class* of evaluation blind spot that current benchmark design cannot express. Functional test suites for hate speech detection stratify test cases by target group and by attack type, and multilingual suites add a language axis; none provides an axis for register variation *within* a language. For a diglossic language this is not a missing row in a table but a missing dimension, and a detector can score well on every existing axis while failing completely on this one. Systems evaluated only along the available axes will report accuracy figures that do not describe their behaviour in deployment.

Second, it supplies a construction method for benchmark data that does not exist in the wild. The standard assumption in corpus-based NLP is that a phenomenon can be found by filtering enough text; we show a case where the phenomenon is absent from the collectable channel for structural sociolinguistic reasons, so that no amount of filtering will surface it. Controlled generation plus native validation is a general answer to this class of problem, applicable wherever a linguistically real phenomenon is systematically excluded from the data-collection surface — code-switched varieties, honorific and politeness systems, and formal registers of any language whose public digital use skews informal.

Third, the diagnostic instrument itself is reusable engineering. The register-pragmatic matrix, the generation and validation protocol, and the multi-detector probe form a template that can be re-instantiated for another language by substituting the register taxonomy and the target-group list, at a cost of a few hundred model calls. For practitioners building or auditing moderation systems, the practical implication is concrete: an evaluation set that does not stratify by register will overstate a detector's coverage on any diglossic language, and the overstatement will be largest for exactly the most socially damaging attacks.

4.6. Limitations

(1) **Native inter-rater reliability is low, not confirmatory.** Section 3.4 reports the full multi-validator diagnosis: per-validator authenticity rates of 55% (Mukhlis), 91% (Yekti), and 45% (Daniel); pairwise Krippendorff's α of 0.095 (Mukhlis–Yekti), 0.779 (Mukhlis–Daniel), and -0.039 (Yekti–Daniel); and a 3-rater α of 0.336, below the conventional 0.667 threshold used even for tentative conclusions [37]. The disagreement decomposes into an instrument artifact (harmonizing the judgment recovers $\alpha = 0.519$ for Mukhlis–Yekti) and a genuine sociolinguistic difference in how much code-switching counts as "authentic Javanese." The 55%/97%/11% figures in Table 4 should be read as one qualified evaluator's estimate, not an inter-subjectively confirmed ground truth; future work with this instrument should report a per-¹⁰ter range rather than a single pooled percentage. (2) **Advisory scale.** 108 examples across 36 cells is sufficient for a controlled proof of concept, not a training dataset; the synthetic set demonstrates the failure mode rather than benchmarking it at statistical power. (3) **Generator bias.** DeepSeek's krama examples use a small set of recurring devices (the "lacks-isin/unggah-ungguh" accusation in N3b; "Mugi...enggal" opener in N2); whether these devices are representative of real krama hate diversity or artifacts of LLM text distribution is an open empirical question. (4) **Regional krama.** DeepSeek defaults to Central-Javanese krama prestige norms; East-Java/Arek krama variants are under-represented or absent. The same prestige-purity-vs-heterogeneous-urban axis reappears on the validation side (Section 4.3: validators' differing tolerance for code-switched Javanese as "authentic"), suggesting this is a general property of the register, not an artifact of one generator model. (5) **Detection scope.** The probe covers five detector models, all using the same cultural prompt and taxonomy (Section 2.8); they are independent as vendors and checkpoints, not as judgment criteria, so this is not evidence against a differently-prompted or differently-trained detector. (6) **Target-category construct validity in the all-detector-evading subset.** Of the 9 cells that evade all 5 detectors

(Section 3.5), the JELAS_HATE judgment (Section 2.7) shows that only 5 are described by both Yekti and Daniel as genuinely ambiguous irony targeting a SARA identity axis (Section 4.2); the other 4 evade for reasons unrelated to pragmatic blindness — 3 target *politik_kolektif*, which both validators independently read as political cynicism rather than identity-directed hate, and 1 (intra-Java *Arek-vs-Mataraman*) leaves the attacked party unstated. This suggests *politik_kolektif* may not be a construct comparable to the ethnic/religious/gender target categories for this niche, and that at least one N3a stimulus under-specifies its target. We disclose this rather than fold these 4 cells into the blind-spot claim (Section 4.2); future iterations of the diagnostic suite should either sharpen the *politik_kolektif* construction or treat it as a distinct category from SARA-identity targets.

Taken together, these six limitations map directly onto a research agenda, and we state that mapping explicitly rather than leaving it implicit. Limitations (1) and (3) — rater disagreement and generator formulaicity — call for a larger validation panel drawn from multiple Javanese speech communities, scoring a stimulus set produced by several generators, so that authenticity is estimated as a distribution rather than as one evaluator's judgment. Limitations (2) and (5) — advisory scale and shared-prompt detection scope — call for extending the probe to fine-tuned classifiers and deployed commercial moderation APIs on a larger stimulus set, which would establish whether the blind spot is a property of prompted LLMs specifically or of hate speech detection as currently practised. Limitations (4) and (6) — Central-Javanese bias and target-category construct validity — call for regionally stratified generation covering East-Java and Banyumasan *krama* norms, and for a sharper operational boundary between identity-directed hate and political derogation. The same agenda transfers to any other diglossic language whose formal register is absent from public digital text.

4.7. Ethics and dual-use

This work is a detection audit, not a generation showcase: LLMs are used to construct the minimum set of stimuli needed to demonstrate a detection failure mode, and the paper's contribution is the diagnosis — a taxonomy and a measured blind spot — not the stimuli themselves. All synthetic texts were generated as controlled research stimuli, excluded from the public repository, and handled under an ethics policy that treats them as equivalent to expert-produced test stimuli. The labeling corpus consists of public social-media posts; handles, mentions, and contact information were anonymized before labeling and the clean release is the only public version.

Dual-use statement. A method that reveals which forms of hate speech evade automated detection could, in principle, be repurposed to craft evasive hate speech. We mitigate this risk in three ways. First, the released artifact is the *diagnosis* — taxonomy, codebook, detection results, and evaluation scripts — not a general-purpose generation model. ¹³ fine-tuned or otherwise repurposable generator is released. Second, the synthetic stimulus set itself **will be released under a restricted, gated, research-only license** (no redistribution, request-based access), following norms established by adversarial test-suite resources such as HateCheck [31]. Third, the paper's conclusion (Section 5) is a call to close the blind spot in moderation systems, not to exploit it; we regard this as analogous to responsible vulnerability disclosure in security research, where documenting a flaw is the step that precedes and motivates its repair.

5. CONCLUSION

We diagnosed a systematic detection blind spot in Javanese hate speech: ironic and coldly contemptuous hostility carried in the *krama* honorific register evades automated detection almost entirely, while explicit *ngoko* hate is caught without difficulty by every model tested. To make the blind spot measurable despite *krama* hate being structurally absent from social media — a collection paradox confirmed by two independent scarcity measurements — we constructed a register-stratified diagnostic

stimulus set, using large language models as a construction method and validating authenticity with three native and near-native raters. The failure traces to pragmatic implicature (*pasemon*) rather than to surface politeness: polite but non-ironic derogation is still detected at high rates, so it is the ironic inversion, not the honorific vocabulary, that defeats the detectors. Cheaper local models fail at *krama* altogether, producing Indonesian instead of Javanese. Human raters converge on the same uncertainty as the machines, which suggests the difficulty is a property of deniable indirect speech itself and not solely a deficiency of current models.

The contribution to informatics is methodological rather than merely descriptive. Existing functional test suites for hate speech detection stratify by target group, attack type, and language, but not by register within a language; for a diglossic language that omission hides an entire class of failure, and a detector can pass every available axis of evaluation while missing the most status-damaging attacks in the language. This paper supplies the missing axis, together with a way to populate it when the phenomenon cannot be collected from public text at all. The practical implication for moderation-system design is concrete: an evaluation set that does not stratify by register will overstate a detector's real coverage on any diglossic language. The register-pragmatic taxonomy, diagnostic construction scripts, native validation instrument, statistical analysis code, and detection probe are released as reproducible infrastructure — the stimulus set itself under a restricted, research-only licence. Future research should extend the probe beyond prompted LLMs to fine-tuned classifiers and deployed commercial moderation APIs, in order to establish whether the blind spot is specific to zero-shot prompting or general to hate speech detection as currently practised; should widen native validation to a larger, regionally stratified panel so that authenticity is estimated as a distribution rather than as one evaluator's judgment; and should re-instantiate the taxonomy for other diglossic and honorific-rich languages, in Indonesia and beyond, to test whether register-borne hostility is a general blind spot of multilingual content moderation.

2 CONFLICT OF INTEREST

The authors declare that there is no conflict of interest between the authors or with the research object in this paper.

GENERATIVE AI DISCLOSURE

In accordance with the journal's generative artificial intelligence policy: large language models are the *object of study and primary method* of this work [45], [46], [47] and are described in full in Sections 2, 3, and 4. Separately, the authors used an AI assistant to help draft and copy-edit portions of this manuscript. All AI-assisted text, all numerical results, and all references were reviewed and verified by the authors, who take full responsibility for the content; no references were generated without author verification. No confidential or personally identifiable data were submitted to third-party AI tools beyond the already-public, anonymized corpus.

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